PES University
Department of Computer Applications

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Unit 3 : Computer Vision & Feature Based Approaches

Assignment Questions

1. What is image filtering? Explain the difference between low-pass and high-pass filters.

2. Describe the Sobel edge detection operator and explain how horizontal and vertical gradients are calculated.

3. Explain the steps involved in the Canny edge detection algorithm.

4. Write a Python program to apply Sobel edge detection on a grayscale image and display the result.

5. Write a Python program to apply Canny edge detection on an image.

6. Compare SIFT, SURF, and ORB feature detection methods in terms of speed, accuracy, and applications.

7. Write a Python program to detect ORB keypoints and display them on an image.

8. Explain the difference between similarity-based and discontinuity-based approaches in image segmentation.

9. Define global vs adaptive thresholding with suitable examples.

10. List and explain common problems and limitations faced in image segmentation.

11. What is region growing segmentation? Explain when it is preferred over threshold-based methods.

12. What are graph cuts? Write two applications of graph cut segmentation.

13. Compare object detection and object tracking with suitable examples.

14. Explain the concept of optical flow and how it is used for tracking movement across video frames.

15. Write a Python program to rotate an image by 45 degrees without cropping the content.

16. Write a Python program:

A) crop the center region of an image and display the cropped part.
B) Resize an image to 300 × 300 pixels.
C) Rotate an image without cropping (rotate + auto border adjustment).
D) Apply translation + scaling together on an image.
E) flip an image horizontally and vertically.
F) Convert a color image to grayscale and resize it.